

AI-assisted writing and the impact of ChatGPT on Greek-speaking students with and without learning disabilities. Outcomes from a repeated measures design

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ABSTRACT

This study examines the impact of structured instruction on artificial intelligence (AI) use for improving academic writing in a non-English context. Participants included first-year students enrolled in a teacher education program, including some with disabilities. Three writing tasks were designed: typed essay, AI-based essay, and a critical AI use essay. The repeated measures analyses revealed that targeted AI integration supported overall writing quality, as indicated by improvements in language use, structure, and content. Additionally, students felt more confident when familiarizing themselves with effective prompt strategies. Students with learning difficulties exhibited the most substantial progress, demonstrating enhanced performance and self-efficacy. Unexpectedly, participants perceived typed essays as offering greater equity of access, a finding attributed to limited AI familiarity at the time of the study.

1. Introduction

Large Language Models (LLMs), such as ChatGPT, represent a significant advancement in natural language processing. As a result, since their widespread availability in 2023, the immense capabilities of LLMs have opened up new possibilities for their application in various fields, including education (Dao, 2023; Kasneci et al., 2023). By enabling individualized learning experiences, automating grading and feedback processes, and by supplying instant support to both educators and students, artificial intelligence (AI) can substantially transform teaching and assessment practices (Chen et al., 2020). These possibilities have generated notable excitement among researchers and practitioners who envision that AI-based solutions can strengthen educational outcomes by responding to diverse learner needs (Mustafa et al., 2024). However, despite the considerable optimism associated with the potential of AI, there are also pervasive concerns regarding ethical and moral considerations (Akgun & Greenhow, 2022; Dieterle et al., 2024), the misuse of AI for dishonest practices, such as cheating (Chaudhry et al., 2023; Larsen, 2023; Sullivan et al., 2023), as well as the challenges associated with detecting such occurrences (Kasneci et al., 2023; Swiecki et al.,

2022). These concerns are not unique to AI; during the pandemic-driven shift to emergency remote assessment, university instructors reported heightened challenges with academic integrity and cheating in online exams (Papanastasiou & Solomonidou, 2023). Such cheating and over-reliance on AI technologies could jeopardize students' intellectual growth and preparedness for real-world challenges (van Niekerk et al., 2025).

Recent studies have highlighted the significance of cultivating students' AI literacy skills and competencies in both K-12 and higher education (Casal-Otero et al., 2023; Laupichler et al., 2022). Specifically, AI literacy does not merely refer to students learning how to use AI applications but rather on their ability to learn about and comprehend AI concepts as well as their ethical implications so that they can use AI both effectively and responsibly (Ng et al., 2021). Students' AI literacy requires their basic understanding of how to use computers and digital devices which, in turn, highlights the close relationship between AI literacy and digital literacy (Chiu et al., 2024). Therefore, students' AI literacy in educational settings involves their ability to understand, evaluate, and explain how AI technologies function and their potential effect on society, to collaborate and communicate with AI systems, and

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to use them responsibly and ethically (Chiu et al., 2024; Long & Magerko, 2020).

Studies on AI in education often focus on technological capabilities and ethical debates, leaving a gap in research concerning how educators and students can productively utilize these tools to support learning and the deeper understanding of cognitive long-term outcomes (Mustafa et al., 2024). Moreover, limited attention has been given to developing practical guidelines that equip students with the skills necessary to use AI and other digital tools legitimately (Casal-Otero et al., 2023; Perkins et al., 2024; Pittas et al., 2022). Such guidelines and policies are particularly pertinent for preparing students to participate effectively in a world where AI-driven technologies are becoming widespread (Georgiou et al., 2024; McDonald et al., 2025). Moreover, there is a recognized need to ensure that AI literacy education is widely accessible across diverse cultural and educational contexts, including European settings (Laupichler et al., 2022; Southworth et al., 2023), or countries in which English is not the main language. This is especially important as AI tools are not as efficient in certain languages or contexts in which less training data has become available for training such tools compared to English.

Beyond the cognitive and linguistic demands of composing in a second language, non-native English speakers with learning disabilities often face compounded disadvantages in higher education, where language barriers, such as difficulties with academic vocabulary, idiomatic phrasing, and extended processing time for reading and writing, worsen disability-related challenges in organization, fluency, executive functioning, and text production, resulting in disproportionate cognitive load, lower self-efficacy, and widened performance gaps relative to native English-speaking peers without disabilities. In this context, recent studies have highlighted the potential of generative AI tools to support and enhance learners' writing, specifically for non-native English speakers (Algobaei & Alzain, 2026) and international students (Sterling et al., 2025). Specifically, studies have indicated that generative AI tools can aid both native and non-native English-speaking students with both local (e.g., grammar, syntax, etc.) and global (e.g., coherence, argumentation, etc.) writing issues as well as with organizing ideas and brainstorming (Wang, 2025). Furthermore, studies have also highlighted the potential of generative AI tools to support students with disabilities (SWDs) to overcome the challenges they face in academic writing (Zhao et al., 2025). Specifically, generative AI tools, such as ChatGPT, can act as assistants in writing and reading activities (Pierres et al., 2025) and can positively impact SWDs' writing processes through the explainability, plannability, and expressibility that they offer (Zhao et al., 2026) and increase their argumentation and synthesis skills (Gabay et al., 2026).

Consequently, and to aid in further understanding the implications of generative AI-based learning, this study aims to describe and evaluate a model for incorporating AI literacy in an academic non-English setting. Specifically, it addresses the following questions:

1. How do student views regarding the characteristics of writing assignments that utilize AI change after engaging with AI for the completion of these assignments?
2. To what extent do AI-supported writing tasks appear to narrow the gap between students with disabilities (SWDs) and students without disabilities (SWODs) in perceived task experience and perceived writing quality?
3. To what extent and how does the quality of the assignments submitted by students change after using AI for their completion?

2. Literature review

2.1. AI and language teaching and learning

Advances in AI have introduced a new generation of writing support tools, from smart grammar checkers to sophisticated generative text

assistants. As a result, AI is increasingly being used in language teaching and learning to create intelligent computer-assisted language learning environments that can assist both teachers and students and offer personalized learning, intelligent tutoring, and advanced analytics and evaluation (Huang et al., 2023; Weng & Chiu, 2023). Although the integration of AI in education is still at an early stage, AI tools have demonstrated great potential in transforming and enriching language learning (Hockly, 2023). In this context, generative AI is gaining considerable ground with tools such as ChatGPT being more widely used by education stakeholders (Law, 2024).

AI-powered writing tools have become increasingly available to support students in various aspects of the writing process, ranging from smart grammar checkers to sophisticated generative text assistants. As a result, the proper use of such tools can yield significant benefits (De la Vall & Araya, 2023). For example, research has shown that the integration of AI into language teaching and learning activities results in an increase in students' language learning motivation, enjoyment, and engagement (Yuan & Liu, 2025). Additionally, when using AI tools and robots, students' cognitive engagement, communication skills, and empathy in the context of language learning can be positively affected (Derakhshan et al., 2025; Rusmiyanto et al., 2023).

AI-mediated language activities can effectively improve students' language and speaking skills through tools that can scaffold the learning process by serving as peer learners, providing recommendations, and offering automatic and real-time feedback (Bin-Hady et al., 2023; Fathi et al., 2024; Weng & Chiu, 2023). Additionally, learners highly value AI writing tools which can influence their writing skills (Lee et al., 2024) and positively affect their overall language learning achievements (Xu & Wang, 2024). In addition to students, teacher trainers and pre-service teachers also express favorable views regarding the adoption of generative AI tools in the subject of English as a foreign language writing due to the benefits they have (e.g., personalized feedback, language support, etc.) but also voice concerns about potential academic misconduct and ethical issues and highlight the need for the reformation of evaluation methods (Söğüt, 2024). Nonetheless, AI tools can support teachers by automating tasks, streamlining the teaching process, allowing teachers to focus more on their students, and enabling them to create innovative teaching approaches (Bonner et al., 2023).

However, there are various factors, such as behavioral intention, self-regulation, motivation, performance expectancy, system quality, digital equity, compensatory technology etc., which should be considered when introducing AI tools to language learning classrooms as they can significantly influence learners' attitudes and participation (Cai et al., 2024). According to Drlić et al. (2025), equity goes beyond just providing material access; it also involves creating cultural and institutional conditions that empower students to use digital technologies effectively. Students should have genuine opportunities to develop the skills and confidence needed to fully engage in academic and social spheres. Digital equity therefore includes access to supportive, compensatory technology, AI-driven language learning approaches as data driven-insights and intelligent tutors, interactive exercises, as well as adaptive learning and personalized feedback, which can be applied to aid students' learning with each affecting students' cognitive load in various ways (Feng, 2025). Ultimately, thoughtful consideration of design and learning strategies is essential for meaningful learning experiences and the pursuit of socially just, sustainable higher education (Martin et al., 2024).

2.2. GenAI-assisted writing in higher education

The research into AI has gained substantial traction in higher education, particularly in the context of student writing tasks, due in part to the rapid technical advancements in this field (Huang et al., 2023). However, relatively few studies have investigated how students and educators integrate AI-generated writing tools in academic settings regarding essay composition, and the attitudes toward such an

integration. In one study, Wang et al. (2024) found that college students in the United States were generally positive about the use of Generative AI tools in their writing assignments. The use of quality prompts that produced writing templates of better quality was among the main strengths cited. On the other hand, they highlighted the lack of personal voice and limited critical thinking as a prominent weakness of AI-generated texts. However, the lack of pre-assessment makes it difficult to measure changes in students' perceptions or writing skills over time. A more robust experimental design could provide deeper insights into the impact of AI tools on learning outcomes. Similarly, Malik et al. (2023) explored Indonesian undergraduate students' views on the use of AI in academic essay writing. The results showed that students had positive attitudes toward the use of AI in writing academic essays, acknowledging that AI improved the quality of their essays in terms of grammar and language use in general. However, it is important to acknowledge that the study relies heavily on self-reported data, which may be subject to bias or inaccuracies from students' subjective perceptions or their limited understanding of the AI tools used. As such, there is a need for further research that employs researcher-administered assessments to more objectively examine the effects of AI-assisted writing. Nguyen et al. (2024) provided further insights into AI-supported academic writing by analyzing 626 AI interactions recorded by ten doctoral students. The researchers concluded that students with high performance demonstrated frequent and purposeful engagement with AI. Their writing, assisted by AI, showed stronger results compared to those with low performance, who tended to use AI in a limited way. As such, high-performing students made more comprehensive use of the tool's capabilities, whereas low-performing students viewed AI as a supplementary resource. However, although this study provided useful insights, it relied on a traditional marking rubric to assess writing performance, which may not entirely reflect the transformative potential of generative AI tools. Future research could benefit from using assessments that can evaluate aspects such as critical thinking or creativity during the writing process.

With reference to how university students use AI for their written assignments, Ćrček and Patekar (2023) found that the majority of students used it for idea generation, paraphrasing, summarizing, and proof reading. As a result, they considered AI as very helpful in their writing process. Again, the reliance on self-reported data introduces the possibility of bias. Along the same lines, Kim et al. (2024) revealed that students used AI as a multi-tasking writing assistant, virtual tutor, and digital peer to support their writing assignments. Regarding AI's impact on academic writing, students acknowledged that GenAI-assisted writing benefited them in the writing process and in their writing performance. Students saw significant improvement in the quality of their writing, reporting improvement in clarity, coherence, organization, grammar, punctuation, and vocabulary. Furthermore, the students benefited from the use of AI in structuring the content and mapping out the process, as well as in developing their metacognitive skills. Most importantly, students valued AI's support in breaking down complex writing tasks, organizing their ideas in a logical order, and spotting language errors and inconsistencies. Nevertheless, since the study lacked objective methods for measuring changes in students' academic writing, future studies could use pre- and post-task evaluations conducted by independent assessors. In a related investigation, Zhang and Wu (2024) reported that a group of first-year postgraduate students experienced notable improvement in both the organization and language elements of writing. Texts generated with AI support tended to be more coherent, demonstrated richer vocabulary, and contained fewer spelling errors.

Systematic reviews have further illustrated the advantages of AI-supported writing. Khalifa and Albadawy (2024) concluded that AI aided academic writing by assisting with idea generation, with the structuring of content, with the synthesis of information, as well as for maintaining clarity, conciseness, and coherence. Likewise, in a review of 30 studies, Imran and Almusharraf (2023) identified that AI provided notable support in pinpointing grammar and syntax errors, structuring

text, and ensuring coherent content. Overall, university students appear to hold favorable attitudes toward AI-assisted writing and they often benefit from AI's capacity to improve writing mechanics and processes. Nevertheless, additional evidence is required to reach more definitive conclusions regarding the effect of generative AI tools on student writing in higher education, especially from non-English speaking countries. Such practices are especially important in order to increase the diversity of research findings while widening the breadth of representation of humans in research results (Kilanioti et al., 2024; Kim et al., 2024).

Recent work by Kizilcec et al. (2024) adds to this discussion by investigating the anticipated student behaviors from the USA, Australia, and Cyprus in using AI for essay or coding assignments. Based on the results of this study, while academics expected that students might lean on generative AI to create entire essays, students described more situational uses, such as for research purposes, idea generation, revisions, and guidance. However, students and academics also raised concerns about academic integrity while recognizing the opportunities for AI to enhance learning and increase efficiency in completing tasks. These findings underscore the evolving conversation around AI's role in university writing assessments and the need to explore these issues in practice.

2.3. AI and students with disabilities in higher education

Although AI has the capability to support learners with disabilities (Zhang et al., 2024) and create inclusive language learning environments (Anis, 2023), a recent systematic review of the literature has identified that only 2% of the studies analyzed have focused on the topic of special education (Mustafa et al., 2024). In one such study, Pierrès et al. (2024) revealed that SWD perceived ChatGPT as an assistant in writing, reading, research, and self-organization. Similarly, Fletcher et al. (2024) found that SWDs saw ChatGPT as an assistant for writing, reading, research, and organizational tasks. Similarly, Fletcher et al. (2024) noted that a small proportion of SWDs relied on GenAI as a virtual tutor and used it mainly to clarify concepts, offer suggestions, and summarize relevant material. However, this study does not explore the perspectives of faculty or other stakeholders, which could provide a more holistic view. Kalniņa et al. (2024) compared pre-service teachers with and without disabilities, focusing on the benefits and difficulties of AI. While students with disabilities identified advantages, including support for language-related barriers, they also expressed concerns over potential reduction in critical thinking and the risk of plagiarism. Results indicated that only a small fraction of participants used AI consistently in their studies, pointing to a gap between the recognized possibilities of AI and its actual adoption by future teachers. This shortcoming raises concerns, given the importance of AI literacy for teachers who will educate the next generation of students. However, the design does not address change over time or causality; it captures perceptions and self-reported use only.

AI's role in enhancing accessibility for SWDs in higher education has also been documented in some recent studies. Sain (2024) found that according to educators', students', and policymakers' views, AI significantly improves accessibility for SWDs including those with physical, hearing or vision impairments. As stated by the participants, AI-driven applications support students with visual impairments as they have the potential to interpret words and magnify images for boosting comprehension. Following the participants' feedback, machine learning-based applications have the potential to convert text into sign language, thus, educational access is achieved for students with hearing impairments. Finally, AI supports students with physical or locomotor impairments by attending lectures from home. On a similar note, Osorio et al. (2024) showed that the combination of artificial intelligence, simulation, and e-collaborative (AISEC) tools have the potential to support accessibility for SWDs in higher education, especially in analytics courses. For instance, AI tools can provide support for students with language barriers through personalized learning, simulation tools

facilitate understanding through modeling real-world processes, and e-collaboration tools to enable instance communication among learners and instructors. Nevertheless, there is no mention of a formal pretest being conducted to assess participants' baseline knowledge or skills before the interventions, leaving some uncertainty about the initial competencies of the students involved.

3. Methods

3.1. Sample of the study

The current study involved 26 first-year students enrolled in the teacher education program at a European university. Participants were recruited during the second week of classes through classroom announcements. All students enrolled in the class were invited to participate in the study, and all consented, resulting in a 100% participation rate. Prior to recruitment, all necessary ethical approvals were obtained from the National Bioethics Committee (EEBK EIT 2024.01.276). Written informed consent was obtained from all participants in accordance with ethical guidelines.

Of the 26 students, five were identified as having learning difficulties. The sample was predominantly female, comprising 21 female students (80.8%) and five male students (19.2%). The mean age of the participants was 18.38 years ($SD = .75$). Seventeen students (65.4%) were enrolled in the Primary Teacher Education program, while the remaining nine students (34.6%) were enrolled in the Pre-Primary Education program. All participants were native speakers of Greek, which is the language of instruction in the current university.

The adequacy of the sample size was examined by using the G*Power 3.1 calculator (Faul et al., 2007). By setting the significance level of .05, a power of .95, and a large effect size (.4), the analysis identified that the minimum sample size required was equal to 18, which verified that our sample of 26 did meet the requirements for our repeated measures analyses.

3.2. Procedures and context

In the context of a core first-year course titled Academic Literacy, which aims to equip students with essential skills crucial for their success throughout their four-year undergraduate studies, a time-series experimental study was conducted. This study took place within a dedicated 3-h workshop session designed to discuss, implement, and examine the utilization of AI in writing assignments. The workshop was centred around a specific writing task that students were required to complete under three different conditions (Typed essay, AI assisted essay, Critical use of AI for essay writing). The writing task was based on the following prompt: "What type of skills and knowledge do you expect to gain through your studies (in Pre-primary or Primary Education) to equip you with for education in the year 2040?"

The workshop's design focused on three writing tasks based on the same writing prompt. For the first writing task (Typed essay), students were allotted 20 min to compose their responses to the writing prompt, using laptops without Wi-Fi access to ensure independent brainstorming and composition. Upon completion, they submitted their typed paragraphs to their instructor.

Before the explanation of the second writing task (AI assisted essay), the students were introduced to ChatGPT as a tool for enhancing their written texts. This was an essential step in the writing process, as 3 students (11.5% of the sample) had indicated that had never used AI in the past. Therefore, after the brief introduction to ChatGPT the students were instructed to create without any guidance, their own prompts in ChatGPT intended to improve the initial essays that they had previously submitted. Upon completion, the students sent their prompts along with the ChatGPT output to their instructor.

For the third writing task (Critical use of AI for essay), the students were first asked to engage in paired discussions regarding their use of

prompts, as well as to analyze both versions of their submitted writing tasks in order to evaluate the improvements made to their texts by AI. Subsequently, in groups of 5-6 students, they collectively evaluated the prompts used by each group member to determine which was the most well-articulated, accurate, and comprehensive, while providing justifications for their selections. Following the group discussions, the students collaborated to synthesize a new prompt that embodied all the identified characteristics of an effective prompt. Finally, they applied this refined prompt to further improve the AI-enhanced versions of their texts. These final prompts along with their outputs were again sent to the instructor.

The workshop concluded with a plenary session where the most effective prompts from each group were shared and discussed. These prompts were written on a whiteboard to facilitate a group discussion focusing on the key characteristics that contribute to an effective prompt. The discussion also explored the role of AI in enhancing students' texts and the extent to which it contributes to text improvement, fostering a critical reflection on the integration of AI tools in academic writing practices.

3.3. Data collection instruments and analyses

During the workshop, a questionnaire was administered to the all class participants. The questionnaire had three sections. The first section included some demographic information, along with a few questions regarding student's prior experience with AI use. The second section included questions aimed at obtaining student's perceptions regarding the writing tasks that they engaged in, based on the Huber et al. (2023) framework for online assessments, which emerged from the synthesis of key design considerations that are essential for the development of effective online assessments. The Huber et al. (2023) framework was selected because it captures six dimensions that are central to digitally mediated assessment and are especially relevant when AI is integrated into writing tasks: academic integrity, student experience, authenticity, information integrity, feedback, and equity in access. Since the aim of the study was to compare students' perceptions of different assessment conditions rather than attitudes toward AI in isolation, this framework provided an appropriate conceptual basis for item construction. Based on this framework, six questions were asked on a 7-point Likert scale (1 = not at all, 7 = very much):

1. To what extent do you believe that this writing task ensures academic integrity by preventing copying and other forms of cheating/inappropriate assistance? (*Academic Integrity*)
2. To what extent do you believe that this writing task creates a positive experience among students, so they do not become anxious? (*Student Experience*)
3. To what extent do you believe that this writing task requires the use of skills that correspond to the actions you will need to perform in your future workplace? (*Authenticity*)
4. To what extent do you believe that this writing task protects your personal data and privacy? (*Information Integrity*)
5. To what extent do you believe that this writing task allows for the provision of timely feedback for the purpose of improving your abilities? (*Feedback*)
6. To what extent do you believe that this writing task is it flexible enough to be completed by anyone (e.g., by individuals with any kind of difficulties)? (*Equity in Access*)

In the third section of the questionnaire, students were asked to evaluate the degree to which they evaluated the easiness of the activity (1 = very difficult, 10 = very easy), the degree to which they liked the activity (1 = not at all, 7 = very much), as well as the grade that they believed that their writing outputs were worth (out of 10). The second and third sections of the questionnaire were administered three times, once after the completion of each of the three writing tasks in order to

obtain the student opinions regarding each of the three writing tasks.

The study used a mixed methods approach to answer the relevant research questions. In regard to the quantitative data analyses, a series of one-way repeated measures ANOVA (with three timepoints) were performed to evaluate the change in opinions regarding the three assessment types that were completed in class (see Fig. 1). Post hoc analyses were also performed to identify where the differences occurred. In regard to the qualitative data, the submitted writing outputs of the three writing tasks by the students were evaluated separately by two researchers with expertise in teaching writing in the language of instruction. The evaluations were performed by using rubrics. The categories that were included in the rubric concerned a) *Language Use* (50%) - focusing on aspects such as style and expression, syntax, and spelling; b) *Structure and Organization* (25%) - focusing on aspects such as logical and sequential structure, as well as formatting, and layout; and c) *Ideas* (25%) - focusing on aspects such as justification and documentation, clarity of the developed argument, and originality of the content.

To determine the interrater reliability of the evaluations, a rating of "1" was first given to the total scores for each writing task that the researchers were in agreement with, and a rating of "0" to the total scores of the writing tasks that there was disagreement on. To calculate the interrater reliability of the assigned grades, the agreement scores were summed up and then divided by the total number of writing tasks. Finally, the interrater reliability was calculated by dividing the agreement scores by the total number of students, and multiplying it by 100. The overall interrater reliability, measured as percent agreement, equalled 78%.

4. Results

In order to identify the variations in students' opinions regarding the three writing tasks, one-way repeated measure analyses were performed. However, the assumption of sphericity first had to be tested by using Mauchly's Test of Sphericity test. As shown in Table 1, the assumption for sphericity was violated in regard to Academic integrity ($W = .664$, $\chi^2 = 9.813$, $p = .007$) and Student experience ($W = .444$, $\chi^2 = 19.465$, $p = .000$). Therefore, the Greenhouse-Geisser correction was applied for these violations.

The results of the repeated measures ANOVA (Table 2) revealed that statistically significant differences were identified only in Equity in access, with a marginal significance for Student experience. The descriptive statistics, presented in Table 3, show that regarding Equity in access, students believed that the typed essay that did not require the use of AI provided the most amount of Equity of access ($M = 6.08$, $SD = 1.09$)

Table 1

Assumption of Sphericity results.

Within subjects effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon ^b	
					Greenhouse-Geisser	Huynh-Feldt
Academic integrity	.664	9.813	2	.007	.749	.786
Student experience	.444	19.465	2	.000	.643	.663
Authenticity	.799	5.371	2	.068	.833	.885
Information integrity	.919	1.954	2	.377	.925	.999
Feedback	.883	2.851	2	.240	.896	.963
Equity in access	.977	.553	2	.758	.978	1.000

among the writing tasks. Conversely, they considered the Critical use of AI as the task with the least amount of Equity of access ($M = 5.46$, $SD = 1.27$) among the three tasks. The post hoc analysis revealed that the statistically significant differences occurred between the tasks of Typed essay and AI assisted essay ($p = .026$).

In contrast, the results for Student experience were in the opposite direction. The students considered the Critical use of AI as the task that was most positive in this perspective ($M = 5.73$, $SD = 1.22$) while they regarded the typed essay as the task that was least positive in this perspective ($M = 4.81$, $SD = 1.39$). For this characteristic, the statistically significant variations occurred between the Typed essay and the Critical use of AI task ($p = .028$).

The second research question compared the variations between SWDs and SWODs regarding their experiences with the three writing tasks. Since the class included only five SWDs, only descriptive statistics were used for this comparison. When comparing the responses of the two groups of students in regard to the typed essay, some large differences were identified along three dimensions (see Table 4). First, the SWDs perceived the typed essay task as the one providing the least positive experience for the students in regard to their anxiety ($M = 3.60$, $SD = 2.07$) as opposed to SWODs ($M = 5.10$, $SD = 1.04$). However, the SWDs perceived that this writing task was more suitable for obtaining timely feedback ($M = 6.40$, $SD = 1.34$) compared to their peers ($M = 5.19$, $SD = 1.25$). Overall, the SWD were much more positive regarding the other two writing tasks that involved the use of AI compared to the first task where they had to type their essay on their own with no help from AI.

Regarding the second writing task that involved the AI assisted essay, the responses of the two groups were quite similar. The only variation

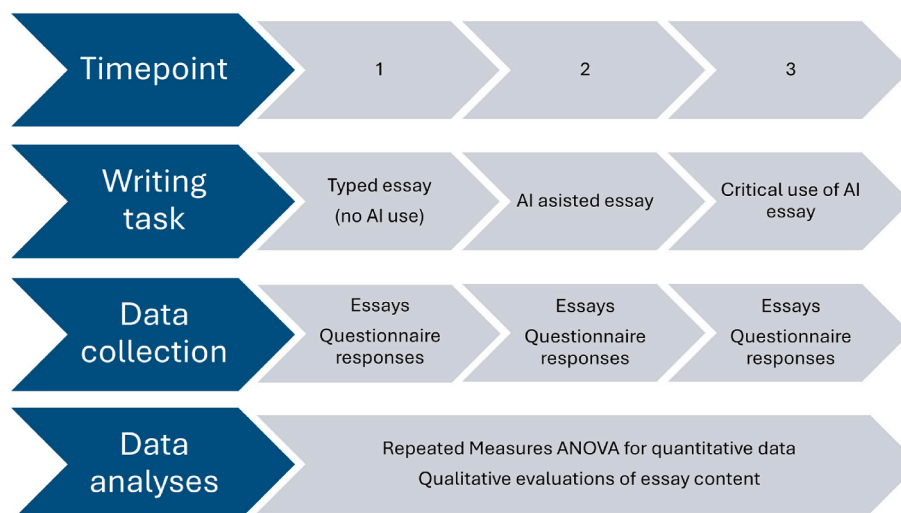


Fig. 1. Study design overview.

Table 2
Repeated-measure ANOVA results.

Measure	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared	Observed Power ^a
Academic Integrity	2.846	1.497	1.901	.775	.433	.030	.156
Student experience	11.077	1.286	8.616	3.367	.066	.119	.482
Authenticity	3.256	2	1.628	1.626	.207	.061	.327
Information integrity	5.120	2	2.560	2.659	.080	.100	.503
Feedback	.320	2	.160	.184	.832	.008	.077
Equity in access	10.692	2	5.346	4.776	.013	.160	.770

Table 3
Descriptive statistics on student perceptions of three assessment types.

	Typed essay	AI assisted essay	Critical use of AI for essay	Typed essay	AI assisted essay	Critical use of AI for essay
	Mean			SD		
Academic Integrity	4.19	3.77	4.15	1.88	1.88	1.46
Student experience	4.81	5.27	5.73	1.39	1.71	1.22
Authenticity	5.27	4.77	5.04	1.19	1.68	1.46
Information integrity	5.38	5.04	4.77	1.2	1.37	1.61
Feedback	5.42	5.28	5.42	1.33	1.24	1.27
Equity in access	6.08	5.69	5.46	1.09	1.49	1.56

Table 4
Descriptive statistic differences among SWDs and SWODs.

		Typed essay		AI assisted essay		Critical use of AI for essay	
		SWDs	SWODs	SWDs	SWODs	SWDs	SWODs
Mean	Academic Integrity	4.00	4.24	4.40	3.62	5.00	3.95
	Student experience	3.60	5.10	5.40	5.24	5.40	5.81
	Authenticity	5.40	5.24	5.60	4.57	5.40	4.95
	Information integrity	6.00	5.24	6.00	4.86	5.60	4.57
	Feedback	6.40	5.19	5.25	5.29	5.60	5.38
	Equity in access	5.40	5.14	5.40	5.76	5.80	5.38
SD	Academic Integrity	2.83	1.67	2.41	1.77	1.87	1.32
	Student experience	2.07	1.04	2.30	1.61	1.67	1.12
	Authenticity	1.82	1.04	1.95	1.60	1.67	1.43
	Information integrity	1.22	1.18	1.41	1.31	1.67	1.57
	Feedback	1.34	1.25	2.06	1.10	1.67	1.20
	Equity in access	1.82	1.20	2.30	1.30	1.79	1.53

occurred on the aspect of Information integrity, where the SWDs ($M = 6.00$, $SD = 1.41$) considered that the AI written task ensured their personal data and privacy to a larger extent compared to their peers ($M = 4.86$, $SD = 1.31$). This same pattern was also identified in the Critical AI use task.

In regard to the evaluation of the Critical use of AI (writing task 3), a large difference was observed that had to do with the characteristic of academic integrity among the two groups of students. More specifically, the SWDs considered that this task ensured academic integrity by preventing cheating and inappropriate assistance to a much larger extent

($M = 5.00$, $SD = 1.87$) compared to SWODs ($M = 3.95$, $SD = 1.32$). The same pattern was observed in regard to Integrity of information where the SWD believed that this writing task protected their information integrity to a larger extent compared to SWOD.

When looking across the three writing tasks, the responses of the SWODs did not show large variations indicating that without disabilities, one may be able to handle a variety of writing tasks and writing conditions. However, large variations were identified among the SWDs, the largest of which, had to do with Student experience since they considered the typed task (with no AI use) as the least favorable among

the three with a difference of 1.80 on a 7-point scale. The second largest difference was identified again among the students with learning difficulties since they believed that the Typed writing task was the most suitable for obtaining timely feedback.

The aim of the third research question was to evaluate the quality of the student's writings based on the three writing tasks. After analysing students' written essay scores provided by their instructors, the overall results showed that AI benefited the students' writing outputs. The average scores obtained for the first writing task was equal to 6.05 (SD = 1.63), compared to an average of 7.52 (SD = 1.33) on the second writing task and an average of 7.89 (SD = 1.12) on the third. The results were further broken down in terms of language use, structure and organization, and ideas. Regarding language use, the student scores on this aspect of their writing were equal to 5.61 (SD = 1.51), 8.45 (SD = .66), and 9.13 (SD = .70) respectively, indicating that there were considerably improvements in the quality of students' essays regarding style and expression, grammar, syntax, and spelling. With regard to structure and organization, AI helped students to produce essays with a logical and sequential structure since the average scores for this category went from 5.92 (SD = .45) on task 1, 7.91 (SD = .27) on task 2, and 8.21 (SD = .65) on task 3. In reference to ideas, the average scores went from 6.19 (SD = .88) on task 1, 7.20 (SD = .96) on task 2, and 7.49 (SD = 1.50) on task 3. Finally, the students were also asked to indicate the degree to which they believed that AI was able to help them improve their writing. The SWDs reported a mean usefulness rating of $M = 7.50$, (SD = 1.73), whereas SWODs had a mean rating of $M = 5.95$, (SD = 1.23).

When evaluating their own typed essays (writing task 1), SWDs assigned themselves lower scores compared to SWODs. On a scale from 0 to 10, the mean self-evaluation score for SWDs was $M = 4.40$, (SD = 2.41), while SWODs had a mean score of $M = 6.14$, (SD = 1.28) (Table 5). This pattern was also reflected in their actual grades, as provided by their instructors. The instructors graded the typed essays of SWDs with a mean of 4.14 (SD = 1.20), and the SWODs with a mean of 6.50 (SD = 1.39). In writing task 3 though (Critical use of AI), this gap stops to exist as the SWDs self-assessed themselves with an average grade of 8.60 (SD = 1.67) compared to the average of 8.38 (SD = 1.36) of the SWODs. When comparing their actual grades across writing tasks, the SWDs increased from 4.14 (SD = 1.20) on writing task 1, to 7.77 (SD = .54) on writing task 3. In contrast, the SWODs increased their scores from 6.50 (SD = 1.39) to 7.91 (SD = 1.20).

The overall correlation between students' self-assessed grades and their actual grades on the typed essay was equal to .415 ($p = .035$), indicating that to a large extent, the students were aware of their actual writing level on the specific task. In contrast, the correlations between the self-assessed and actual grades for the other two tasks were not statistically significant, indicating that the students were not as able to

obtain a realistic view of the quality of their AI outputs in the current tasks. For the AI assisted essay (task 2), the correlation was equal to $-.04$ ($p = .854$), and for writing task 3, it was equal to $-.23$ ($p = .320$). When looking at the student assignments in a qualitative manner, the SWDs students demonstrated notable weaknesses in orthographic accuracy, spelling, and syntactic conciseness in their initial short essays. Their writing was often characterized by long asyndetic sequences of subordinate volitional clauses, typically introduced by the particle "na", resulting in reduced syntactic cohesion and limited variation. Moreover, the absence of an explicit governing clause, led to structurally incomplete or loosely organized sentences. In addition, weaknesses in style, lexical choice and phrasal precision were observed. Although some of these issues also appeared in the essays of other students, they were more pronounced in the writing of SWDs. Within this context, the linguistic improvements observed in the revised essays through AI assistance can be classified into several categories.

First, orthographic corrections were identified, particularly spelling errors and incorrect use of accents in very common and frequently repeated words. For example, the incorrect form "σχόλια" (comments) was used instead of "σχολεία" (schools). This mistake is particularly significant because the incorrect spelling corresponds to another existing Greek word, which means "comments", thereby increasing the potential for confusion for the reader or examiner.

Second, sentence restructuring, the use of linking words and increased informational density were observed. While students tended to produce longer sentences with less organized content, AI-generated revisions resulted in shorter sentences enriched with linking words and more specific information. For example, instead of a single sentence containing five volitional clauses, the AI produced three shorter main clauses with clearer structure.

Third, lexical and phrasal precision was improved through more appropriate word choice. For instance, "to discover several kinds of talent" was reformulated as "to discover and cultivate their personal talents", suggesting greater specificity. Similarly, the verb phrase "face any situation" was revised to "handle every situation", conveying a more active and practical engagement with the scenario described.

Fourth, the introduction of adverbs contributed to greater precision in expression. For example, "to be able to teach at schools" was reformulated as "to be able to teach effectively at schools", which conveys the intended meaning more concisely.

Finally, stylistic refinement was evident in several cases. For example, "to know the right ways to teach" was reformulated as "we need to acquire knowledge of appropriate teaching practices". This shift from verb-based, informal phrasing to a more nominalised-oriented expression reflects a more precise and formal style appropriate for academic discourse.

Table 5
Assigned and self-assessed grades from the three writing tasks.

		Typed essay		AI assisted essay		Critical use of AI for essay	
		SWDs	SWODs	SWDs	SWODs	SWDs	SWODs
Mean	Self-assessed grade	4.40	6.14	8.20	7.86	8.60	8.38
	Actual grades	4.14	6.50	6.51	7.74	7.77	7.91
SD	Self-assessed grade	2.41	1.28	1.92	1.74	1.67	1.36
	Actual grades	1.20	1.39	1.61	1.20	0.54	1.20

Despite the improvements that were evident in the writing tasks with the use of AI, significant language issues were observed due to the fact that ChatGPT is not trained as well in Greek as it is in English. Therefore, its Greek language capabilities are not yet fully developed. When students asked ChatGPT to improve their text in terms of style, grammar, and syntax, some problems were observed in the final outputs of tasks 2 and 3. These included “clumsiness” in the Greek language, particularly in morphology, style, and the use of appropriate verbs or words, which resulted in a deduction of marks. For example, consider the phrase “*ὅταν οι φοιτητές εκτίθενται σε διδασκόντες*” (student 24). When students intended this to mean “when the students are associated with lecturers” ChatGPT edited it in a way that used a verb with a double meaning in Greek. The chosen verb implies that the students were being insulted in front of the lecturers, which is not appropriate in this context. The verb “*Προσδοκῶ*” is a more sophisticated verb that means *anticipate* and is used in the Christian Orthodox Creed (“*προσδοκῶ ἀνάστασι νεκρῶν*” meaning *I expect the resurrection of the dead*). Due to its religious and formal connotation, a Greek native speaker would not typically choose this verb as a synonym for *expect*. A notable morphological mistake was also identified in the phrase “*Αποτελεσματικούς μεθόδους*”. The noun *μεθόδους* (meaning *methods*) is feminine, but it often incorrectly takes a masculine article and adjective to its ending -ous, which is the same as that of some masculine nouns. As a result, ChatGPT erroneously applied a masculine adjective ending (*αποτελεσματικούς*), instead of the correct feminine form (*αποτελεσματικές*). Such grammatical inaccuracies not only reflect poorly on the students’ command of the language but also can lead to confusion and misinterpretation of the content.

These language issues highlight the critical need for careful evaluation and editing when utilizing AI tools like ChatGPT for academic writing, especially in languages where the AI’s proficiency is limited. Relying solely on AI-generated revisions without thorough proofreading can result in the inclusion of errors that affect the overall quality and professionalism of academic work. It emphasizes the importance of developing students’ language skills alongside their ability to effectively use technology. Educators and students must be aware of these limitations and approach AI assistance as a supplementary tool rather than a definitive solution, ensuring that the final output meets the rigorous standards of academic writing.

5. Discussion

The present study offers insights into how targeted instruction in the use of AI in academic settings may enhance writing quality and student engagement, particularly, among individuals who encounter challenges in traditional assessment scenarios, and how students evaluate such variations among writing tasks. Overall, the results showed that the quality of the assignments improved with the proper use of AI, as indicated by both the students’ assigned grades as well as their self-assessed grades for their writings. This result is in line with those from studies carried out in other educational settings that found that the use of AI improves students’ writing output (Kim et al., 2024; Wang et al., 2024) and academic performance (Kong et al., 2021). The present study added to these findings by including a pre-assessment to track changes in students’ perceptions and writing skills over time, offering clearer insights into how AI tools affect learning outcomes through a carefully designed experiment. Unlike earlier work relying on traditional rubrics (e.g., Nguyen et al., 2024), in this study, independent assessors evaluated student writing for qualities such as critical thinking and creativity. Among the 3 writing tasks that the students engaged in, the one in which they had to prepare a collaborative AI prompt was the most promising one, first in terms of the quality of the written outputs and second in terms of providing a more positive student experience while alleviating anxiety in regard to the completion of the assignment. This result, along with findings from previous studies (see Southworth et al., 2023), demonstrates the effectiveness of the integration of AI in the learning process for providing dynamic educational experiences and

personalized learning. The benefit was especially prominent for SWDs, since the AI integrated tasks enabled them to overcome many of the challenges that they faced in regard to writing assignments.

The responses among the students, regarding their perceptions of their experience with the three writing tasks, identified some unexpected findings. For example, the students considered the Typed essay as the task with the most amount of equity of access. This is a surprising result as most of the recent literature on AI supports that AI can increase equity of access, especially due to its ability to enhance personalized education (Maghsudi et al., 2021; Mustafa et al., 2024). A possible explanation for this outcome, however, is that it occurred due to the inexperience of some students in the use of AI and its potential. As a result, these students might not be aware of its advantages for equity of access. In contrast, they might most likely believe that completing assignments in a traditional manner might be more flexible and more equitable compared to assignments that require the use of specialized software and skills for using AI since some students are less familiar with AI. This inexperience might also be the reason why SWDs believed that they could obtain timely feedback mainly with the typed essays. Although this comes in contrast with multiple studies that focus on the timely feedback that can be provided by AI (see Wang et al., 2024), this is knowledge that the students must have not been familiar with at the time of the study.

The finding that students rated the typed essay as higher in equity of access should not be read as evidence that AI is inherently less equitable. Rather, it suggests that students interpreted “access” in terms of immediate usability and participation requirements. The typed essay demanded fewer prerequisite skills and less technological confidence. By contrast, AI-supported tasks required students to know how to prompt effectively, assess AI suggestions critically, and manage the limitations of Greek-language output. Thus, the results point to a distinction between equity of access and equity of outcomes: the typed task may have felt easier to enter, whereas the AI-supported tasks appeared more effective in reducing performance gaps, particularly for SWDs.

Another surprising finding was that the SWDs considered that the tasks that involved the use of AI ensured their personal data and privacy to a larger extent compared to their peers or to the typed essays. A likely explanation for this result, is that SWDs might feel less exposed to their educators when they can utilize AI to improve their writing. As a result, they might feel that since AI might be able to help them in a way that masks their personal weaknesses and learning disabilities, thus, protecting their personal learning weaknesses. As demonstrated by Deme-triou (2023), the collaborative use of technological tools reduced reliance on more capable peers by offering more balanced learning opportunities and increasing participation for the SWDs. In one instance that was identified, a student with dyslexia was able to overcome some of his learning difficulties through a web-based learning environment, which facilitated better collaboration within his group and, to some extent, camouflaged his potential learning weaknesses in writing. Similarly, in the context of the current study, AI tools may serve as an alternative form of support in writing, offering students with disabilities assistance similar to what they might receive when collaborating with more capable peers, but in a private manner, thereby protecting them from exposure of their personal learning weaknesses.

Regarding academic integrity, a large difference was identified in the Critical use of AI task, since SWDs considered that this task ensured academic integrity by preventing cheating and inappropriate assistance to a larger extent compared to SWODs. One likely explanation for this surprising result is that due to the students’ inadequate training in AI and AI ethics, they were not able to understand how one should be interacting with AI and how such interactions can prevent copying and other forms of cheating/inappropriate assistance. This study does not stand alone and confirms earlier results by Kizilcec et al. (2024) who showed that both students and academics are concerned about academic integrity, yet see AI as a tool to improve learning and efficiency. These results

highlight the ongoing discussion of AI's impact on university writing assessments.

Overall, in respect to the SWDs, the findings indicated that SWDs preferred the use of digital assessment methods compared to traditional ones. For example, the writing tasks that included AI received the highest ratings among SWDs for its support in improving their academic writing which, in turn, also increased their grades in the assigned task. This finding suggests that for SWDs, traditional assessment approaches, such as typed essays, are not among the most effective methods for assessing their writing skills, while utilizing AI appropriately can help them significantly. This result is in line with [Dong's \(2023\)](#) study who also found that AI tools enhanced students' writing skills.

Finally, the Critical AI Use task appeared to empower SWDs because it reduced barriers at multiple levels of the writing process. At the linguistic level, AI supported spelling, accentuation, and lexical precision. At the discourse level, it helped reorganize long and weakly connected sentence sequences into clearer and more cohesive structures. At the cognitive level, the paired and group-based prompt comparison activity externalized planning and revision, thereby reducing executive-function demands and making the revision process more manageable.

5.1. Implications

The results of this study highlight a disparity in the experiences and self-perceptions of SWDs compared to SWODs in the context of writing typed essays. The less positive experience and lower self-evaluation scores among SWDs on the typed essays suggest that traditional written assessments may pose particular challenges for this group, potentially impacting their academic engagement and success. The positive reception of the use of AI for their writing assignments indicates that such technologies can play a significant role in supporting the writing process for SWDs or students who struggle with written communication ([Kim et al., 2024](#)), enhancing their ability to produce quality written work and improving their confidence. These findings suggest that SWDs benefit significantly when they have regular, structured opportunities to use AI tools designed according to their specific learning needs. Instead of treating AI as an optional feature that students might use only if they choose to, instructors can create lesson plans or assignments that require all students, and especially SWDs to practice developing and revising AI prompts. In this way, AI use becomes a central part of the course rather than an afterthought or side activity. This approach also encourages students to work together, sharing feedback on the prompts they create or the text generated by AI. By incorporating peer-review sessions, students learn from one another's successes and challenges, making the process more interactive and supportive. This step-by-step support can help increase SWDs' confidence in their writing and lead to better performance outcomes overall. In turn, the benefits of AI support for students with disabilities may also extend to emotional and psychological domains. Since students who have struggled with writing may experience anxiety or low motivation toward written assignments, AI tools, can provide them with a safety net and on-demand help, which can alleviate some of this pressure.

However, significant work should also be done in order to educate students on the ethical use of AI. Just as they should be provided with opportunities to utilize it appropriately, they should also be provided with information and training on AI literacy in general ([Kong et al., 2021](#)). They need to be made aware of the ethics related to AI use, the biases and misinformation that could occur, as well as issues of academic integrity and information integrity. This is especially important for teacher education programs so that they can shift their conceptualization of "teacher readiness" beyond mastery of foundational pedagogical theories to include topics related to AI literacy as well as cultivating higher order thinking skills ([Kizilcec et al., 2024](#)). This can be achieved through courses that position teacher candidates not merely as users of AI, but as critical analysts who can identify when and how to rely on AI-generated suggestions. Moreover, by focusing on such skills in

teacher education programs, the students themselves could have a model to rely on, regarding how to include AI assistance in their own teaching practice with their own students.

Based on these findings, several practical and policy considerations arise. The integration of AI and universal design for learning (UDL) into higher education necessitates comprehensive teacher training that enhances both technical competence and pedagogical expertise, while also fostering ethical awareness. Teachers need to be supported in translating AI-derived insights into effective instructional methodologies, thereby ensuring that technology facilitates personalized learning, adaptive feedback, multilingual support, and improved accessibility ([Pittas & Nussbaumer, 2025](#)). Furthermore, the establishment of robust institutional frameworks is required to deliver flexible, inclusive, and effective educational experiences tailored to the diverse needs of adult learners; such frameworks should include protocols for data protection, adherence to accessibility standards, and guidelines for harmonizing individualized instruction with inclusive classroom practices ([Mallary et al., 2025](#)). At the policy level, equitable access to AI and UDL depends on ongoing investment in digital infrastructure, funding for inclusive technological resources, and the creation of transparent regulatory mechanisms that ensure accountability and mitigate bias. These initiatives contribute to reducing inequalities by ensuring that all students, regardless of physical or cognitive challenges, have equal opportunities to the learning process. Collectively, these implications delineate concrete steps for educators, institutions, and policymakers to maximize the benefits of GenAI and UDL within adult and continuing higher education.

6. Limitations

Despite providing valuable insights, this study has several limitations that should be acknowledged. First, the characteristics of the sample impose constraints on the generalizability of the findings. The sample size was relatively small, comprising only 26 first-year students from a teacher education program at a European university. Although the G*Power analysis indicated that a minimum sample size of 18 was sufficient for the repeated measures analyses conducted, the small sample limits the statistical power and the robustness of the conclusions. A larger sample would enhance the reliability of the results and allow for more generalizable findings.

Second, the sample was predominantly female, which may introduce gender bias and limit the applicability of the results to a broader population. While the overrepresentation of female participants reflects the gender distribution within the teaching profession, it may not adequately capture the perspectives of male students. Future research should strive for a more balanced gender representation to explore potential differences in perceptions and experiences between male and female students. Moreover, the subgroup of students identified as having learning difficulties was notably small, with only five participants. This small number limits the ability to generalize findings related to students with all types of learning difficulties and reduces the statistical power to detect significant differences between students with and without learning difficulties. This limitation underscores the need to replicate this study on a larger scale to investigate the impact of AI on students with learning disabilities more thoroughly.

It should also be taken into account that this study focused exclusively on first-year teacher education students who were all native speakers of the language of instruction. This homogeneity in language proficiency, which is not English, means that the findings may not be applicable to populations where language barriers could affect the use of AI in writing. This is especially important because academic performance is associated with multiple individual and contextual factors ([Patterson et al., 2003](#)), many of which could not be examined within such a homogeneous sample. Cultural factors and varying degrees of familiarity with AI technologies in different educational contexts were not explored, which limits the generalizability of the results to more

diverse student populations.

Finally, the rapidly evolving nature of AI technologies, particularly in educational settings, means that the interaction between AI and humans can change quickly as both the technologies themselves and their implementations advance. The findings of this study represent a snapshot in time, and longitudinal studies would be beneficial to observe how perceptions and impacts may change as students become more familiar with AI tools and as those tools develop further.

7. Conclusion

In conclusion, this study demonstrates that the integration of generative AI through structured and critically guided instruction significantly enhances academic writing quality and student experiences in higher education. The results show that AI-assisted writing tasks contribute to improved language use, organization, and idea development, with SWDs showing the greatest gains in performance and self-efficacy. Therefore, it can be inferred that generative AI tools could provide real-time linguistic scaffolding, structural guidance, and idea-generation support for students experiencing intersecting linguistic barriers and disability-related challenges in writing, thus enabling them to focus on higher-order thinking and reasoning instead of lower-level transcription demands. At the same time, mixed perceptions regarding equity, academic integrity, and language accuracy highlight the importance of AI literacy, ethical awareness, and contextual sensitivity when integrating such tools into assessment practices. Rather than replacing traditional writing, AI is most effective when positioned as a collaborative and reflective partner that supports cognitive engagement, critical evaluation, and inclusive learning. By embedding AI use within pedagogically sound frameworks that emphasize inclusive practices, higher education institutions can better prepare students—and future educators—to navigate writing and assessment in an increasingly AI-driven landscape.

CRedit authorship contribution statement

Elena C. Papanastasiou: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Evdokia Pittas:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Marina Rodosthenous-Balafa:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Georgios Lampropoulos:** Writing – review & editing, Writing – original draft.

Ethics approval

The study was approved by the Cyprus National Bioethics Committee (EEBK EPI 2023.01/340) as well as by the Ministry of Education, Culture and Sports (07.30.002.002).

Informed consent was obtained from all participants, and their privacy rights were strictly observed. The data can be obtained by sending request e-mails to the corresponding author.

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The authors have no conflict of interest to disclose.

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